



Motivation

RAG pipelines apply the same retrieval strategy to every query regardless of difficulty—wasting calls on easy queries and underinvesting on hard ones. We frame per-query retrieval selection as an **offline contextual bandit**: a lightweight policy routes each query to its best retrieval action, making cost explicit through per-call and rewrite penalties.

- Fixed-action baselines (BM25, dense, hybrid) cannot adapt to query-level variation in difficulty or retrieval confidence.
- A learned router should exploit cheap wins and invest extra calls only when retrieval gain justifies cost.
- Scope: **retrieval-stage reward only**—no LLM fine-tuning, no generated-answer EM/F1.

Contextual-Bandit Formulation

Each query is a one-step routing decision:

$$\pi(s) = \arg \max_a \mathbb{E}[R \mid s, a]$$
$$R = \underbrace{\text{Recall@5} + 0.5 \cdot \text{MRB}}_{\text{retrieval quality}} - \underbrace{c_{\text{rw}} \cdot \mathbf{1}[\text{rewrite}] - c_{\text{call}} \cdot (\text{calls} - 1)}_{\text{explicit cost penalties}}$$

- State s :** lexical query features + BM25 confidence statistics (no LLM at inference time).
- Actions:** BM25 | dense | hybrid | keyword rewrite; Gemini variants are costly optional extensions.
- Policy:** direct-method action-value models trained on full-information reward replay; cross-validated model selection.
- Training:** replayable offline reward table—not online bandit learning; selected-action replay is a diagnostic baseline only.

Implementation Pipeline

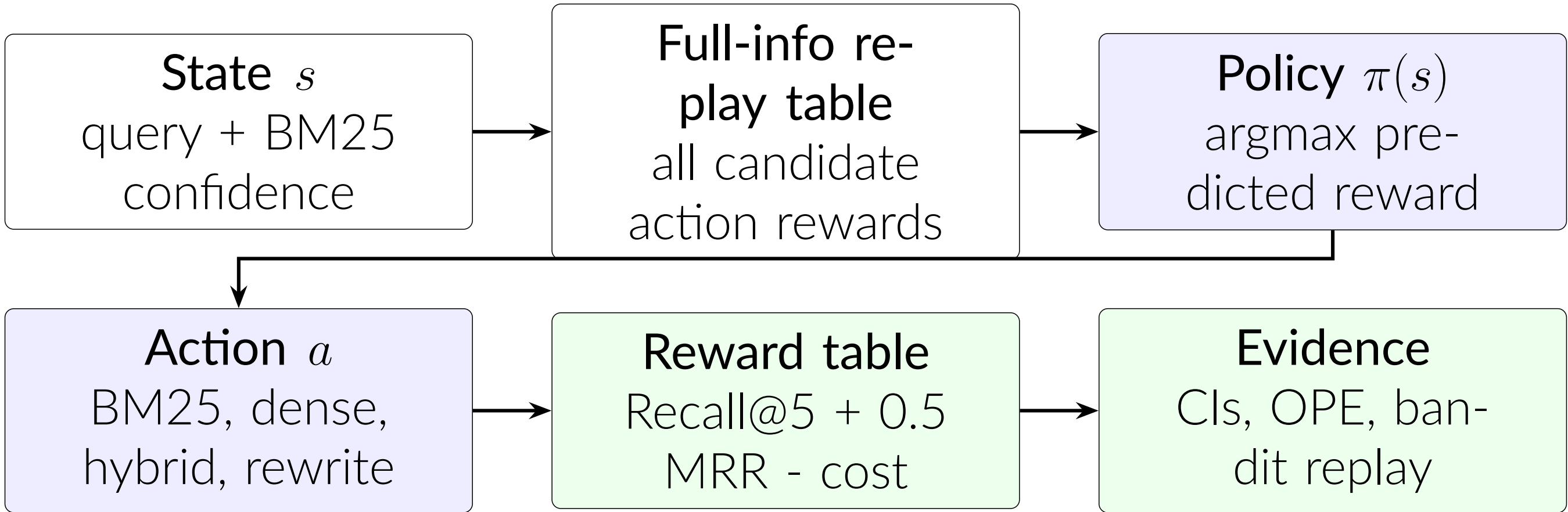
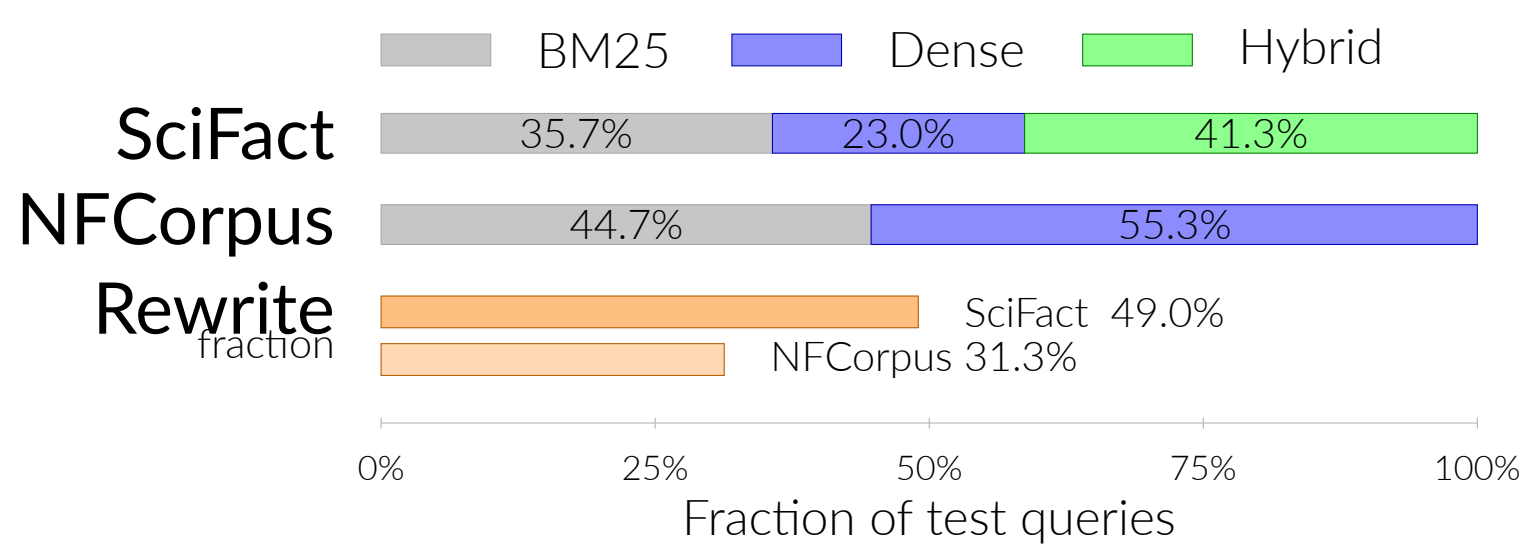


Figure 1. Offline contextual-bandit framing: replayable action rewards train a policy that maps query state to retrieval action.

Policy Action Distribution

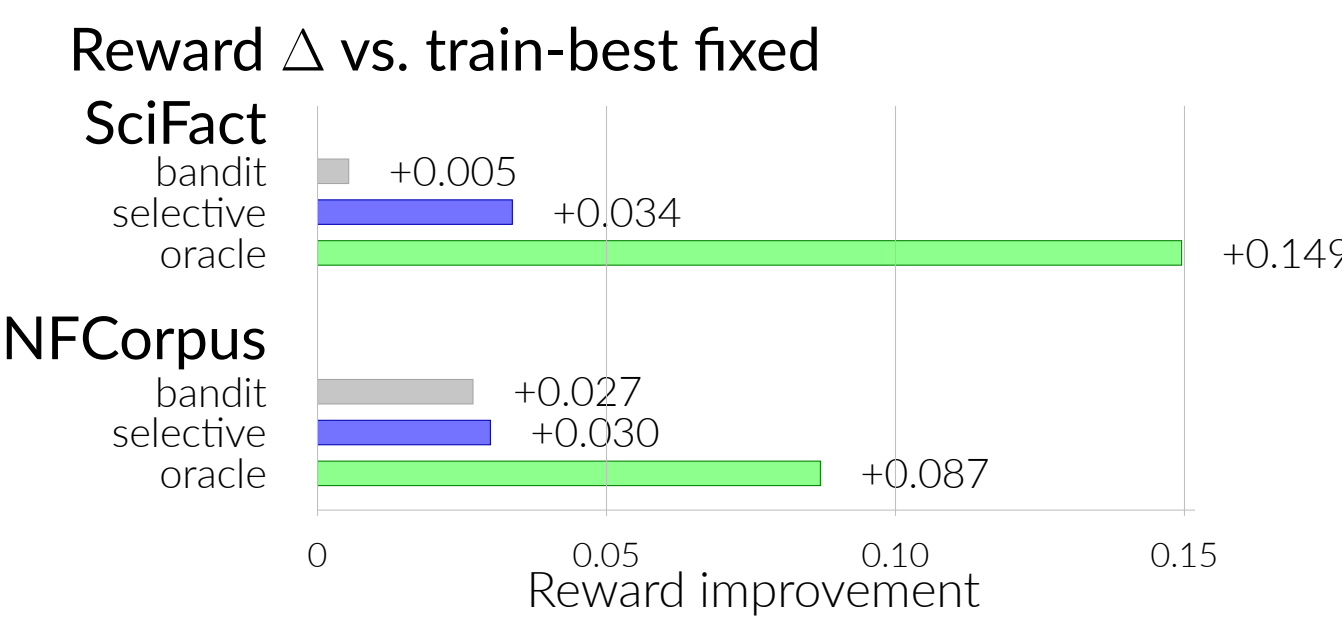


SciFact benefits from hybrid retrieval (41.3%) and frequent rewrites; NFCorpus stays cost-efficient with single BM25 or Dense calls.

Main Results

Dataset	Method	Reward	Δ (95% CI)	Calls/q
SciFact	Train-best fixed	1.057	— (baseline)	2.00
SciFact	Selected-action bandit	1.062	+0.005	1.03
SciFact	Selective policy	1.091	+0.034 [0.009, 0.058]	1.41
SciFact	Oracle upper bound	1.206	+0.150	1.01
NFCorpus	Train-best fixed	0.377	— (baseline)	1.00
NFCorpus	Selected-action bandit	0.404	+0.027	1.14
NFCorpus	Selective policy	0.407	+0.030 [0.005, 0.055]	1.00
NFCorpus	Oracle upper bound	0.464	+0.087	1.02

Reward = Recall@5 + 0.5 · MRR − cost. **Calls/q**: avg. retrieval actions per query (train-best fixed = 2 calls on SciFact). Bootstrap 95% CIs stay above zero on both datasets. Full-corpus BEIR: 600 train / 300 test queries per dataset (SciFact: 5,183 docs; NFCorpus: 3,633 docs).



(a) Reward-delta ladder

Key Results

Metric	SciFact	NFCorpus
Selective Δ	+0.034	+0.030
Oracle gap closed	22.6%	34.4%
Constr. Δ	+0.047	+0.033
Calls/query	1.41	1.00
3-seed win rate	0.667	0.667

above zero on both datasets; constrained utility CIs: [0.014, 0.082] (SciFact) and [0.011, 0.058] (NFCorpus).

(b) Summary

Figure 2. Selective-policy gains with cost-aware and robustness diagnostics.

The selective policy closes **22.6%** of the SciFact oracle gap and **34.4%** of the NFCorpus oracle gap. On NFCorpus, the same reward gain is achieved at **zero extra retrieval cost** (1.00 call/query).

Cross-Dataset Generalization

Routing policies trained on one domain transfer to unseen target domains:

Source	Target	Policy Reward	Target Best	Calls/q
SciFact	SciFact	1.093	1.057	1.48
SciFact	NFCorpus	0.386	0.377	1.60
NFCorpus	NFCorpus	0.407	0.377	1.00
NFCorpus	SciFact	1.064	1.057	1.00

NFCorpus→SciFact transfers with positive reward delta at no extra cost. SciFact→NFCorpus transfers but incurs 1.60 calls/query.

Robustness & Extended Analysis

- Full-info advantage:** direct-method beats selected-action bandit by +0.005 (SciFact) and +0.027 (NFCorpus), confirming the value of full-information training over biased feedback.
- Difficulty buckets:** high action-disagreement queries show larger gains—SciFact +0.049, NFCorpus +0.096—confirming the router targets the queries where routing matters most.
- Budget sweep:** constrained policy ($\lambda = 0.06$) keeps +0.039 utility at 1.05 calls/query on SciFact; NFCorpus is unchanged at 1.00 call/query.
- Gate baseline:** a margin-gating heuristic adds only +0.0006 over selective on SciFact and is no better on NFCorpus—a learned policy outperforms hand-tuned thresholds.
- Three-seed robustness:** win rate 0.667 on both datasets; mean deltas stable across seeds (+0.021 SciFact, +0.025 NFCorpus).

Conclusion

A lightweight contextual bandit can replace fixed-action RAG retrieval, improving retrieval reward on both BEIR datasets while keeping cost explicit. The policy is interpretable (feature-based), transfers across scientific domains, and is strongest on high-difficulty queries where fixed strategies most disagree.

Limitations: retrieval-stage evidence only; training uses full-information replay (not real online logs); OPE is simulated from offline data; semantic features remain analysis-only.

